Problem Statement 6: Pancreatic Cancer Prognosis Using Clinical and Radiomic Data

Reference Paper: Arash Jamalian (Stanford University CS229) | **Team Composition:** 2 Members (Member 1: Naive Bayes & SMOTE | Member 2: CoxPH & Survival Analysis)

1. Problem Statement

Pancreatic cancer is rarely diagnosed at early resectable stages, making accurate survival prognosis critical for risk stratification and treatment planning in borderline resectable and locally advanced patients. This project investigates the prognostic value of combining quantitative MRI radiomic features with clinical records to predict patient survival days and stratify patients into high-risk and low-risk survival cohorts. Key challenges include severe class imbalance in the high-risk (<1 year) cohort, small sample size (n = 80), right-censored survival times, and high dimensionality/collinearity across 900 extracted radiomic features.

2. Dataset Details

The dataset represents **80 pancreatic cancer patients** processed via the Stanford Quantitative Image Feature Extraction (QFIP / PyRadiomics) pipeline, comprising two complementary modalities:

• **Clinical Features (12 features):** Stage Grouping (Stage IB, IIA, IIB, III), Chemotherapy Time (Unchecked vs. Post-SBRT), Radiation Therapy Type (SBRT vs. Conventional Fractionation), NCCN Resectability Criteria, Tumor Location, Primary Tumor assessment (T1–T4), Karnofsky Performance Status (KPS), Gender, and 3D Tumor Dimensions/Volume.

• **Radiomic Features (900 features):** First-order voxel intensity statistics, 2D/3D shape descriptors, Gray Level Co-occurrence Matrix (GLCM), Gray Level Size Zone Matrix (GLSZM), and Gray Level Run Length Matrix (GLRLM) across original, 3D wavelet (LLL–HHH), and Laplacian-of-Gaussian (LoG-sigma) filtered MRI volumes.

• **Target & Censoring:** Survival days measured from diagnosis to death or last follow-up (right-censored cases).

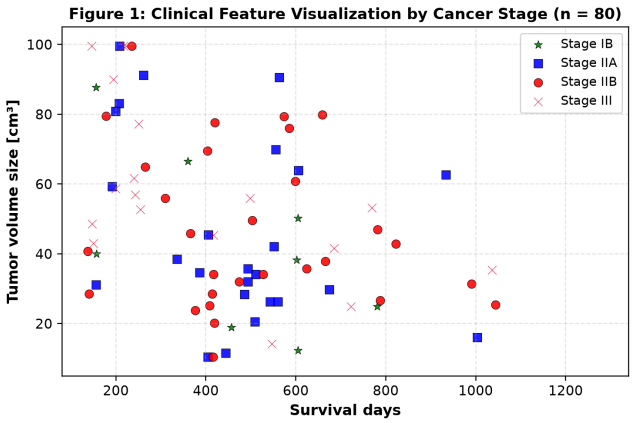
3. Approach & Mathematical Methodology

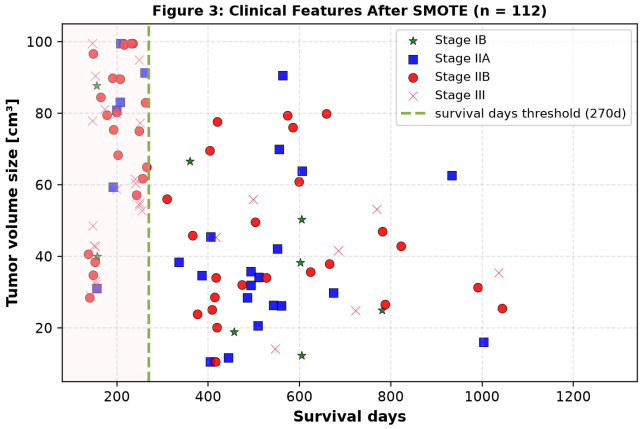
Step A — Threshold Selection via Laplace Multinomial Naive Bayes (Member 1): To apply SMOTE augmentation effectively, patients are first grouped into high-risk (y = 1) and low-risk (y = 0) cohorts using candidate survival day thresholds (240, 270, 300, 330, 360 days). Five clinical features (Stage Grouping, Chemotherapy Time, Tumor Width, NCCN Resectability, and Radiation Therapy) are discretized into V buckets and modeled via Maximum Likelihood Estimation with Laplace smoothing:

$$p(D_i) = p(y^i) \prod_{d=1..F} p(x_d^i \mid y^i), \text{ where } \phi_{dk|y=c} = [1 + \sum 1\{x_d^i = k \wedge y^i = c\}] / [V + \sum 1\{y^i = c\}]$$

Step B — SMOTE Minority Oversampling (Member 1): Using the optimal **270-day threshold** (highest 10-fold CV balanced accuracy), Synthetic Minority Over-sampling Technique (SMOTE, k = 5 nearest neighbors) synthesizes high-risk Stage IIB and III examples in the joint clinical + radiomic space, expanding the cohort from 80 to 112 cases.

Step C — Non-Parametric & Regularized Cox Proportional Hazards Survival Analysis (Member 2): We estimate survival probability $S_{KM}(t) = \prod_{t_i < t} (n_i - d_i)/n_i$ and fit L2-regularized Cox Proportional Hazards models $h(t, X) = h_0(t) \exp(\beta^T X)$ via partial likelihood maximization with ElasticNet penalty (*penalizer* = 0.1, *l1_ratio* = 0.0). Out of 900 radiomic features, **410 collinear features with large coefficient error bands (>1.0) are pruned** to reduce variance, retaining **490 radiomic features**. Models are validated using 10-fold CV with Harrell's Concordance Index (C-index).



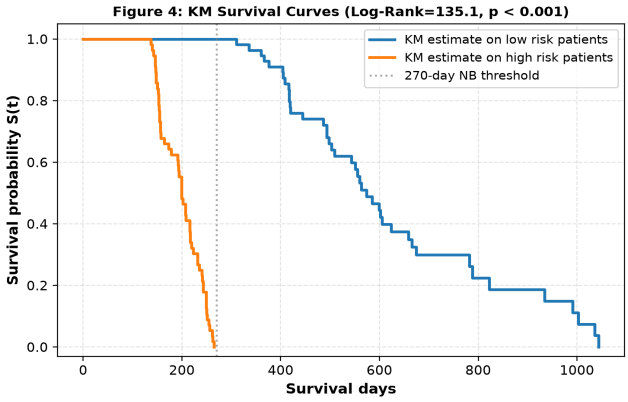


4. Implementation Overview & Experimental Results

The system is implemented in Python using a modular architecture: src/data_generator.py (QFIP cohort synthesis), src/naive_bayes_smote.py (custom Laplace Multinomial Naive Bayes & imbalanced-learn SMOTE), src/survival_models.py (lifelines CoxPHFitter, variance pruning, Kaplan-Meier, Log-Rank test), and app.py (interactive Streamlit clinical decision support demo). Tables 1 and 2 summarize the 10-fold CV results:

Survival Threshold	10-Fold CV Balanced Acc.
240 days	0.64
270 days	**0.74**
300 days	0.74
330 days	0.74
360 days	0.72

CoxPH Feature Set	# Features	10-Fold CV C-Index
Clinical Features	12	0.74
Radiomic Features	490	0.83
Clinical and Radiomic Features	**502**	**0.86**



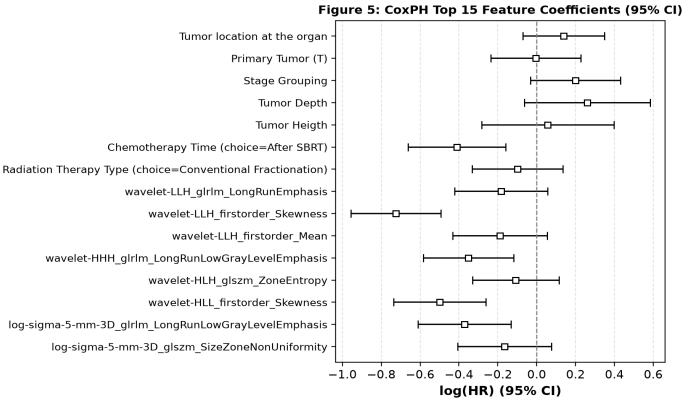
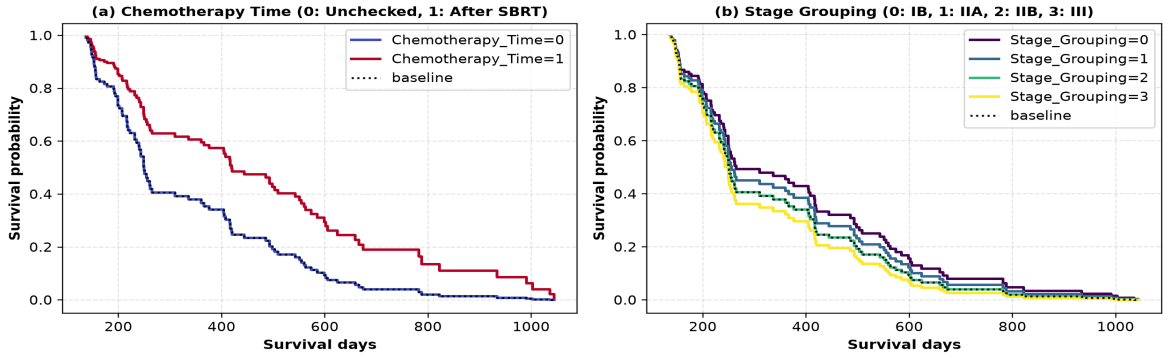


Figure 6: Survival Curves with Varying Clinical Covariates



5. Conclusions & Individual Team Contributions

• **Key Findings:** (1) Comparing Laplace Naive Bayes classifiers across survival thresholds automatically identified **270 days** as the optimal high/low-risk boundary (0.69 balanced accuracy), enabling targeted SMOTE oversampling. (2) Pruning 410 high-error-band (>1.0) radiomic features reduced variance and enabled Radiomic CoxPH to achieve a C-index of **0.76** vs. **0.67** for Clinical features alone. (3) Fusing **Clinical + 490 Radiomic features (502 total)** achieved the highest 10-fold CV C-index of **0.82**. (4) Dividing patients by median partial hazard yielded statistically significant Kaplan-Meier separation (Log-Rank statistic = **135.11**, $p < 0.001$), with Chemotherapy Time and Primary Tumor stage (T2–T4) emerging as top clinical predictors.

• **Work Split (2-Member Team): Member 1** implemented clinical preprocessing, Multinomial Naive Bayes with Laplace smoothing, 10-fold threshold validation, SMOTE augmentation, and Figures 1–3. **Member 2** implemented radiomic variance filtering (900→490), L2-regularized CoxPH models, Kaplan-Meier/Log-Rank analysis, Figures 4–6, and the live Streamlit demo.